

CHINA SOLAR

Expert Call takeaway: Expert expects improving curtailment to drive China demand inflection in 2028

We hosted an expert call to discuss China Solar demand outlook on Jun 22. The speaker has specialized in PV policy analysis and industry research since 2017, serving as a consultant for leading Chinese renewable companies and government agencies.

Key takeaways include: **i)** China's solar installations are projected to decrease by 26% YoY in 2026, with a continued downward trend in 2027 (at a slower pace) due to a diminishing pipeline of new Utility Scale Solar (USS) projects; **ii)** Alleviating grid curtailment is expected to trigger a demand inflection point from 2028 onwards; **iii)** Integration of renewables in 2026-2030 will be supported by multiple structural levers, including policy mandates, increased grid investment, and strong power consumption growth.

Implication to GS coverage: The expert's demand forecast is largely aligned with GSe for 2026 (-25% YoY to 235GW) but notably more conservative for 2027 (forecasting a continued decline vs. GSe of +14% YoY to 268GW). In our view, the extended demand trough implies that the profitability recovery for solar main value chain players may defer to 2028, vs. our current estimate in 2027. That said, a sustained downturn through 2027 could catalyze the exit of more marginal capacity, resulting in a healthier industry landscape over the long term. Within the solar value chain, we maintain our preference for high-efficiency cell and Module players (Buy on Longi) over Glass and Poly players (Sell on Flat Glass A/H, Tongwei, and Daqo A).

Below we summarize our takeaways in detail:

China's solar installation is expected to decline by 26% yoy in 2026: The expert forecasts overall China's solar installation to reach 210–245GW in 2026, or 238GW at the mid point (largely inline with GSe of 235GW). By project type, The expert estimated 125–145GW (135GW at mid point) from Utility Scale Solar (USS) projects, 50–60GW (55GW at mid point) from Distributed C&I projects (DS-C&I), and 45–50GW (57GW at mid point) from Distributed Residential projects (DS-Resi). As compared to GSe, USS demand is largely inline (-18% yoy and -2% vs. GSe of 138GW), while DS-Resi shows unexpected resilience (+3% yoy and +235% vs. GSe of 14GW). Conversely, DS-C&I demand appears weaker than our estimates (-48% yoy and -33% vs. GSe of 83GW).

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- **For DS-Resi**, the expert attributed segment resilience to lower IRR thresholds accepted by POE players amid declining financing costs. Demand is concentrated in regions with robust on-grid volume guarantees and favorable tariff structures, such as Southeastern China.
- **For DS-C&I**, the market is transitioning from grid-connected models to self-consumption driven by Solar+ESS (Energy Storage Systems); this is driven by improved economics with a Solar+ESS LCOE (Rmb0.65-0.7/KWh) now undercut peak electricity prices in most regions. However, larger “direct supply” projects face longer construction cycles due to complex grid-backup requirements, explaining the near-term shortfall versus GSe.

In 2027, China installation may continue a downward trajectory at a milder pace:

The expert anticipates a further, albeit slower, decline due to less new USS project reserve. This trend is unlikely to be fully offset by recovering DS-C&I demand and steady DS-Resi demand. Notably, USS project starts fell 34% YoY in 4M26 following two years of declining approvals. Given the 6–12 month lag between construction and grid connection, these weak starts signal lower USS installation for the upcoming year.

Improving curtailment will drive a demand inflection in 2028 and beyond: The expert noted the low investment sentiment for USS project is mainly due to severe curtailment issue. In 2025, 23 provinces reported a decline in solar utilization hours (averaging -12% YoY), with western regions such as Tibet and Xinjiang seeing the most significant impact. Eastern provinces like Shandong also face integration challenges, as solar output occasionally exceeds 60% of total supply during peak hours.

Looking forward, the expert expects curtailment to improve based on three factors: **i)** Strategic policy shifts to encourage renewable energy consumption, including the NDRC’s inclusion of clean energy consumption as a mandatory provincial KPI; **ii)** Significant grid infrastructure spending, projected to rise to approx Rmb5tn for 2026–30 (vs. Rmb2.77tn in 2020–25); and **iii)** Sustained load growth, evidenced by a 5% YoY increase in electricity consumption in 4M26. The combination of expanded grid capacity and rising demand creates a constructive backdrop for a USS recovery starting in 2028.

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